

Naive Bayes Theorem – Theory and Equation

Naive Bayes is a probabilistic machine learning algorithm used for classification tasks. It is based on Bayes' Theorem and assumes that the features used for classification are independent of each other. Because of this simplifying assumption, it is called 'Naive'. Despite the simplicity of this assumption, Naive Bayes performs very well in many real-world applications such as spam email detection, document classification, and medical diagnosis.

Naive Bayes Equation:

$$P(C | X) = [P(X | C) \times P(C)] / P(X)$$

Explanation of Terms:

- 1 $P(C | X)$: Posterior probability – probability of class C given predictor X.
- 2 $P(X | C)$: Likelihood – probability of predictor X given class C.
- 3 $P(C)$: Prior probability – probability of class before observing the data.
- 4 $P(X)$: Evidence – probability of predictor X.

Theory of Naive Bayes Classifier:

- 1 Naive Bayes classifier is based on Bayes' theorem of conditional probability.
- 2 It assumes that all attributes are conditionally independent given the class label.
- 3 The classifier calculates the probability of each class for a given input instance.
- 4 The class having the highest posterior probability is selected as the predicted class.
- 5 It is widely used in data mining, machine learning, text classification, and spam filtering.

Advantages of Naive Bayes:

- 1 Simple and easy to implement.
- 2 Works well with large datasets.
- 3 Requires small amount of training data.
- 4 Fast prediction and classification.

Applications:

- 1 Email spam filtering.
- 2 Sentiment analysis.
- 3 Document classification.
- 4 Medical diagnosis.

Naive Bayes Classification Example (Han & Kamber)

Naive Bayes is a probabilistic classification algorithm based on Bayes' Theorem. It assumes that all attributes are independent of each other given the class label. This example is commonly presented in the Data Mining textbook by Han & Kamber to demonstrate classification using the Buys_Computer dataset.

Age	Income	Student	Credit_rating	Buys_Computer
<=30	High	No	Fair	No
<=30	High	No	Excellent	No
31..40	High	No	Fair	Yes
>40	Medium	No	Fair	Yes
>40	Low	Yes	Fair	Yes
>40	Low	Yes	Excellent	No
31..40	Low	Yes	Excellent	Yes
<=30	Medium	No	Fair	No
<=30	Low	Yes	Fair	Yes
>40	Medium	Yes	Fair	Yes
<=30	Medium	Yes	Excellent	Yes
31..40	Medium	No	Excellent	Yes
31..40	High	Yes	Fair	Yes
>40	Medium	No	Excellent	No

Problem: Classify a new customer with attributes:
Age <=30, Income = Medium, Student = Yes, Credit_rating = Fair.

We want to predict whether the customer will buy a computer.

Step 1: Calculate Prior Probabilities

Total records = 14

Buys_Computer = Yes → 9

Buys_Computer = No → 5

$P(\text{Yes}) = 9/14$

$P(\text{No}) = 5/14$

Step 2: Likelihood Probabilities

For class YES:

$P(\text{Age} \leq 30 \mid \text{Yes}) = 2/9$

$P(\text{Income} = \text{Medium} \mid \text{Yes}) = 4/9$

$P(\text{Student} = \text{Yes} \mid \text{Yes}) = 6/9$

$P(\text{Credit} = \text{Fair} \mid \text{Yes}) = 6/9$

For class NO:

$P(\text{Age} \leq 30 \mid \text{No}) = 3/5$
 $P(\text{Income} = \text{Medium} \mid \text{No}) = 2/5$
 $P(\text{Student} = \text{Yes} \mid \text{No}) = 1/5$
 $P(\text{Credit} = \text{Fair} \mid \text{No}) = 2/5$

Step 3: Posterior Probability

For YES:

$$P(X|\text{Yes})P(\text{Yes}) = (2/9 \times 4/9 \times 6/9 \times 6/9) \times (9/14)$$

For NO:

$$P(X|\text{No})P(\text{No}) = (3/5 \times 2/5 \times 1/5 \times 2/5) \times (5/14)$$

The class with higher probability becomes the predicted class.

Final Classification:

The probability for class YES is higher than NO. Therefore, the Naive Bayes classifier predicts that the customer **WILL BUY A COMPUTER (Yes)**.

Bayesian Classification – Detailed Example

Bayesian classification is a probabilistic classification technique based on Bayes' Theorem. It predicts the class label of a data sample by calculating probabilities using prior knowledge from training data. This method is widely used in machine learning, data mining, spam filtering, and medical diagnosis.

Bayes' Theorem:

$$P(C|X) = (P(X|C) \times P(C)) / P(X)$$

Where:

$P(C|X)$ = Posterior probability (probability of class C given data X)

$P(X|C)$ = Likelihood (probability of observing X if class is C)

$P(C)$ = Prior probability of class C

$P(X)$ = Evidence probability

Example Problem:

We want to predict whether a student will PASS or FAIL in an exam based on study hours (High or Low). We use previously observed training data.

Student	Study Hours	Result
S1	High	Pass
S2	High	Pass
S3	Low	Fail
S4	Low	Fail
S5	High	Pass

Step 1: Calculate Prior Probabilities

Total students = 5

Number of Pass = 3 $\rightarrow P(\text{Pass}) = 3/5 = 0.6$

Number of Fail = 2 $\rightarrow P(\text{Fail}) = 2/5 = 0.4$

Step 2: Calculate Likelihood

For students who studied HIGH hours:

$P(\text{High} | \text{Pass}) = 3/3 = 1$

$P(\text{High} | \text{Fail}) = 0/2 = 0$

Step 3: Classify New Student

Suppose a new student studies HIGH hours.

Posterior probability for PASS:

$$P(\text{Pass}|\text{High}) \propto P(\text{High}|\text{Pass}) \times P(\text{Pass})$$

$$= 1 \times 0.6 = 0.6$$

Posterior probability for FAIL:

$$P(\text{Fail}|\text{High}) \propto P(\text{High}|\text{Fail}) \times P(\text{Fail})$$

$$= 0 \times 0.4 = 0$$

Step 4: Final Decision

Since $0.6 > 0$, the Bayesian classifier predicts that the new student will PASS.